

ChE 210

Homework Set #2

Due: Friday, 12 September 2025

(1) A chemical reaction $A \rightarrow B$ is carried out in a closed vessel. The following data are taken for the concentration of A, $C_A(\text{g/L})$, as a function of time, $t(\text{min})$, from the start of the reaction:

$t(\text{min})$	0	36	65	100	160	∞
$C_A(\text{g/L})$	0.1823	0.1453	0.1216	0.1025	0.0795	0.0495

A proposed reaction mechanism predicts that C_A and t should be related by the expression

$$C_A = C_A^0 - k t$$

$\ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -kt$
where k is a reaction rate constant, and the zero and infinity subscripts on the concentrations indicate the initial and very long time concentration values.
(a) Do the data support this prediction? If so, determine the value of k .
(b) If the tank volume is 50.5 gallons and there is no B in the tank at $t=0$, how much B in grams does the tank contain in two hours?

(2) (a) Find two different ways to linearize the equation $r = A z / (B + z)$ where A and B are adjustable constants. The reaction rate is r , and the concentration is z . Use both linearizations to fit the data to the right using perpendicular offsets. Compare the values of the constants for the two linearizations and the quality Q^2 of the fits.
(b) Use the solver on Excel to find a numerical solution for fitting the nonlinear equation to the data using vertical offsets. You will need a reasonable first guess for the solver in order to get the numerical method to converge. Use one of your results in part (a) for this guess. Compare this result with the two results in part (a).

(3) Murphy 1.12
(4) Murphy 2.11 (1 ton = 2000 lbm)

(5) A mixture of ethanol and water contains 60 wt% water.
(a) Assuming volume additivity of the components, estimate the specific gravity of the mixture at 20°C. What volume (in liters) of this mixture is required to provide 150 moles of ethanol?
(b) Repeat part (a) with the additional information that the specific gravity of the mixture at 20°C is 0.93518 (making it unnecessary to assume volume additivity). What percentage error for the volume of the mixture arises from the volume additivity assumption?

P2.11 In 1970, the year the Clean Air Act was passed by the U.S. Congress, the average hydrocarbon emission was about 14 grams per mile traveled. That number dropped to about 1 gram per mile by the year 2000. The total number of motor vehicle miles traveled increased in the same period from about 1100 billion miles in 1970 to about 2600 billion miles in 2000. Calculate the total hydrocarbon emissions in 1970 and in 2000, in tons. What is the percent increase or decrease over the 30-yr period?

1970 period:

$$14 \frac{g}{mile} \cdot 1100,000,000 \text{ mile} = 1.54 \times 10^9 \text{ g hydrocarbon} \cdot \frac{1 \text{ lbm}}{453.6 \text{ g}} \cdot \frac{1 \text{ ton}}{2000 \text{ lbm}} = 1697.5 \text{ tons hydrocarbon}$$

2000:

$$1 \frac{g}{mile} \cdot 2600,000,000 \text{ mile} = 2.6 \times 10^9 \text{ g hydrocarbon} \cdot \frac{1 \text{ lbm}}{453.6 \text{ g}} \cdot \frac{1 \text{ ton}}{2000 \text{ lbm}} = 2.910 \text{ tons hydrocarbon}$$

$$\frac{(2600) - (1.54 \times 10^9)}{(1.54 \times 10^9)} \cdot 100 = -93.1\%$$

Off by a factor of 10^3 , but overall mostly correct, -5 pts

5)

4) 40 wt% ethanol $P_{eth} = 0.289 \text{ g/cm}^3$
40 g ethanol $P_{water} = 1 \text{ g/cm}^3$

$$V_L = \frac{60 \text{ g}}{1 \text{ g/cm}^3} = 60 \text{ cm}^3$$

$$V_L = \frac{40 \text{ g}}{0.789 \text{ g/cm}^3} = 50.69$$

$$V_{tot} = 110.697$$
$$P_{mix} = \frac{100 \text{ g}}{110.697} = 0.903 \text{ g/mL} = SG$$
$$P_{L=1}$$

$$150 \text{ mol ethanol} \cdot \frac{46.08 \text{ g}}{1 \text{ mol}} = 6912 \text{ g} \cdot \frac{1}{46} = 17280 \text{ g mix}$$

$$SG \cdot P_{tot} = \frac{m \cdot x \text{ g}}{V_{tot} \text{ mix}}$$

$$V_{tot} = \frac{m \cdot x \text{ g}}{SG \cdot P_{tot}} = \frac{17280}{0.903 \cdot 1}$$

$$V_{tot} = 19136 \text{ mL} = 19.136 \text{ L}$$

$$150 \text{ mol} \cdot \frac{46.08 \text{ g}}{1 \text{ mol}} = 6912 \text{ g ethanol}$$

$$V_{tot} = \frac{17280}{0.90316 \cdot 1} = 19177 \text{ mL}$$

$$V_{tot} = 19.177 \text{ L}$$

$$\% \text{ error} = \frac{19177 - 19136}{19136} \cdot 100$$

$$\% \text{ error} = 0.21\%$$

$$20/20$$

1) a) Solve for k and see if its constant

$$t=36 \quad k = -\frac{1}{36} \ln \left(\frac{0.1453 - 0.0495}{0.1823 - 0.0495} \right) \quad t=65 \quad k_{65} = 0.0094$$
$$t=100 \quad k_{100} = 0.0092 \quad t=160 \quad k_{160} = 0.0092$$

the data does support ✓
The $k_{avg} = 0.0092$

b)

$$C_A(z) = (C_{A0} - C_{A\infty}) e^{-kt} + C_{A\infty}$$
$$= (0.1823 - 0.0495) e^{-0.0092 \cdot 120} + 0.0495 = 0.0935$$

$$C_B = C_{A0} - C_A \quad M_B = C_B \cdot \left(30.5 \text{ gal} \cdot \frac{3.785 \text{ L}}{1} \right)$$
$$C_B = 0.0944 \quad M_B = 10.25 \text{ g}$$

2) a) method 1: $x = z$
 $y = \frac{z}{r}$

$$r = \frac{Az}{B+z}$$
$$r(B+z) = Az$$
$$rB + rz = Az$$
$$rB = Az - rz$$
$$rB = z(A-r)$$
$$r \left(\frac{B}{A-r} \right) = z$$

$$r = \frac{A}{B} z + A$$

$$\frac{1}{\text{coeff}} = A \quad B = \frac{1}{\text{coeff}} \cdot \text{int}$$
$$A = \frac{1}{0.006} = 166.7$$
$$B = 18.45$$
$$R^2 = 0.9996$$

$$A = \frac{1}{\text{int}} \quad B = \text{coeff} \cdot A$$
$$A = \frac{1}{0.0094} = 106.38$$
$$B = 0.1002 \cdot 0.1002 = 0.01004$$

P1.12. In the case study, we evaluated the economics of using either benzene or glucose as a raw material for making adipic acid or catechol. Prices of raw materials can change, sometimes dramatically, as supply and demand fluctuate. Suppose the price of all raw materials stays the same except glucose. How much would the price of glucose (\$/kg) have to drop for the glucose-to-adipic acid process to be economically competitive with the benzene-to-adipic acid process? What about for the catechol process?

Glucose - adipic: Table 1.7 and 1.8

$$-86,880 \cdot X + 770 \cdot 0.2 + 28,100 \cdot 1.51 = 27,100$$
$$X = 0.20 \text{ \$/kg}$$

The Price would have to drop to 0.2 to be equal with the benzene process, and drop further to be economically competitive

Glucose - Catechol: Table 1.11 and 1.8

$$-86,770 \cdot X + 21,150 \cdot 0.462 = 89,300$$
$$X = 0.10 \text{ \$/kg}$$

The Price would have to drop below 0.10 $\frac{\$}{kg}$ to be economically competitive with the benzene process

method 2:

$$r = \frac{Az}{B+z}$$
$$\frac{1}{r} = \frac{B}{Az} + \frac{z}{Az}$$
$$\frac{1}{r} = \frac{B}{Az} + \frac{1}{A}$$
$$r = \frac{A}{B} z + A$$

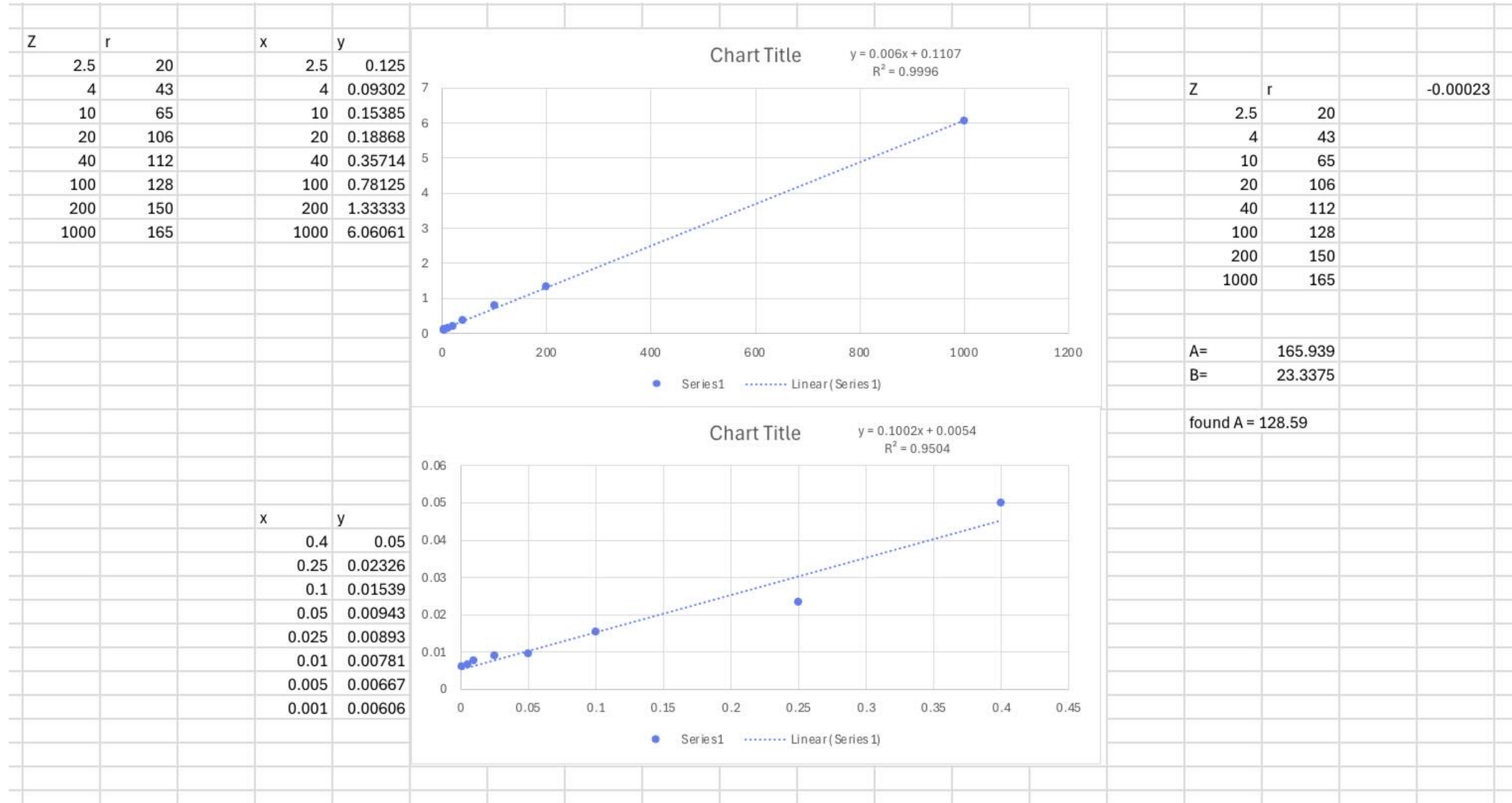
$$\frac{1}{r} = \frac{B}{Az} + \frac{z}{Az}$$

$$\frac{1}{r} = \frac{B}{Az} + \frac{1}{A}$$

$$r = \frac{A}{B} z + A$$

$$r = \frac{A}{B} z + A$$

its clear the first one is closer than the other one using the R^2 value
 $0.9996 > 0.9504$



$$20/20$$